SUJAY ARJUNAN

EDUCATION

PROGRAM	INSTITUTION	%/CGPA	COMPLETION
MTech, Biomedical Engg	Indian Institute of Technology Madras	--	2026-2028
BE, Biomedical Engg	Sri Sivasubramaniya Nadar College of Engineering	7.614	2022-2026
XII	Velammal (Main)	94.16%	2022
X	Velammal (Main)	86.0%	2020

SCHOLASTIC ACHIEVEMENTS

- Co-authored paper titled "Electroencephalographic Comparison of Fatigue-Induced Cognitive Effects Across Age Groups " under peer review for the Journal of Mechanics in Medicine and Biology (JMMB)
- Co-authored paper titled "Deep Learning Architectures and Use of Explainable AI for Model Predictions in Classifying Ear Disorders" under peer review for the Journal of Mechanics in Medicine and Biology (JMMB)

PROFESSIONAL EXPERIENCE

Product Development Recruit - Innotrat Labs
May 2025 - July 2025

- Worked as a product development recruit to aid in software development for STM32 based products
- Aided in development of a vision chatbot in classifying tools and instruments
- Worked with SIM800 module in establishing bilateral communication

Hospital Trainee - Apollo Hospitals
December 2025

- Assisted in daily inspections across departments , preventive maintenance and minor repair of equipments
- Observed endoscopy and MRI, CT procedures and obtained comprehensive inference
- Observed how labs work in clinical settings
- Learnt how standards and regulations are followed

PROJECTS

Deep Learning Model for Classifying images of Ear Disorders
December 2025

- Built a CNN model to classify diseased tympanic membrane images from healthy ones with 91% accuracy
- Also built 3 other CNN models to compare differences.
- Paper for this project accepted for submission to the Journal of Mechanics in Medicine and Biology (JMMB)

Electroencephalographic Comparison of Fatigue-Induced Cognitive Effects Across Age Group
July 2025 - March 2026

- Worked on EEG data acquisition and signal processing to find age-related differences in cognition in pre and post fatigue conditions
- Identified significant results between groups based on bandpower and ERS/ERD
- Paper titled the same accepted for submission to the Journal of Mechanics in Medicine and Biology (JMMB)

Pain Quantification using EMG signals with ML
December 2023 - July 2026

- Worked on EMG signals to identify feature trends for different levels of pain
- Built an Ensemble Machine Learning model to predict 5 classes of pain with 82.5% accuracy

SKILLS

Technologies: Matlab, HTML, CSS, Python SQL, OpenSim, Machine Learning, Neural Networks, Signal Processing, STM32
Frameworks & Libraries: MySQL, Sci-kit Learn, Pandas, Numpy, tensorflow
Practices & Project Management: Frontend and backend development, Model training & validation
Languages: English, Tamil (native), Hindi, Japanese (current N1)

POSITIONS OF RESPONSIBILITY

Head, SSN Gaming Club, 2024-2026	Conducted three flagship events during my tenure as head Collaborated with other institutes to organise events
Event Head, Invente 2026	Conducted Paper Presentation Event in Invente 2026, SSN where students were allowed to showcase their research work to experienced veterans and bag a prize.
Member, SSN MUN	Participated in various MUNs with one of the best teams in Tamilnadu

OTHER HIGHLIGHTS

- Managing a hosted domain in its entirety called 'The Tangent' where complex concepts are explained in a friendly way. Link: The Tangent — The part of the lecture nobody talks about
- Won the SLCU MUN 2024 as the 'Best Delegation'
- Completed three NPTEL SWAYAM Courses: Conservation Economics, Classics in Neuroscience and Electronic Systems for Cancer Diagnosis
- Completed Advanced Diploma in Python Programming with an A+